Matematiska Institutionen
Department of Mathematics

Examensarbete

Steady periodic water waves solutions using asymptotic approach

Shahid Hasnain

Reg Nr: LiTH-MAI-EX--2011/14-SE
Linköping 2011



Linköpings universitet
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Mathematics

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LiTH-MAI-EX--2011/14-SE

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Linköping, 13 June, 2011

Avdelning, Institution
Division, Department

Division of Applied mathematics
Department of Mathematics
Linköpings universitet
SE-581 83 Linköping, Sweden

Datum
Date

2011-006-13

Språk
Language

☐ Svenska/Swedish
☒ Engelska/English
☐ _____

Rapporttyp
Report category

☐ Licentiatavhandling
☒ Examensarbete
☐ C-uppsats
☐ D-uppsats
☐ Övrig rapport
☐ _____

ISBN
__

ISRN
LiTH-MAI-EX--2011/14-SE

Serietitel och serienummer **ISSN**
Title of series, numbering __

URL för elektronisk version
http://www.mai.liu.se
http://www.ep.liu.se

Titel
Title

Steady periodic water waves solutions using asymptotic approach

Författare
Author

Shahid Hasnain

Sammanfattning
Abstract

The aim of this work is to study the relation between two invariants of water flow in a channel of finite depth. The first invariant is the height of the water wave and the second one is the flow force. We restrict ourselves to water waves of small amplitude. Using asymptotic technique together with the method of separation of variables, we construct all water waves of small amplitude which are parameterized by a small parameter. Then we demonstrate numerically that the flow force depends monotonically on the height.

Nyckelord
Keywords

Steady two dimensional water wave, perturbation approach, separation of variable, pressure head and flow force.

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Symbols

$\eta(x)$	Free surface.
ϕ	Potential function .
ψ	Stream function .
$\bigtriangledown$	Del Operator
g	Gravitational acceleration.
Q	Flux
$\mathbb{R}$	Pressure head.
S	Force flow
$\bigtriangleup$	Laplace Operator
ϵ	A small parameter
L	Period
$\mathbb{X}$	New variable

Abbreviations

BVP	Boundary value problem
ODE	Ordinary differential equation

Acknowledgments

First of all I would like to thank The Almighty Allah who is the great source for knowledge and wisdom, who gave me strength and conviction to understand and complete this project.
I would like to show my gratitude to my Professor Vladimir Kozlov, thank you very much for providing me with much needed advice during this project, being patient, answering all the silly questions. I would like to thank Prof.Lars Elden who started my journey of computational mathematics when I came to Sweden. I would like to thank Henrik Branden, to teach me the numerical approach of mathematics. I would like to thank Prof.Iryna Yakymenko who teach me important courses to understand the physics of the problem and also she is my program coordinator, my hearty thanks to her. I would like to thank my program Chairman Sven Stafstrom. I would also like to thank the Swedish educational system, not only for the great educational experience and learning but also for allowing me to experience the Swedish life style. I would like to thank my parents for unconditional love and support. Without your encouragement this work would not have been possible. Thanks for being there when i needed you. Last but not least, I would like to thank all my friends especially Touqeer Pasha and Syed Asmat Shah for their kind support throughout the project.

Abstract

The aim of this work is to study the relation between two invariants of water flow in a channel of finite depth. The first invariant is the height of the water wave and the second one is the flow force. We restrict ourselves to water waves of small amplitude. Using asymptotic technique together with the method of separation of variables, we construct all water waves of small amplitude which are parameterized by a small parameter. Then we demonstrate numerically that the flow force depends monotonically on the height.
Keywords:
Steady two dimensional water wave, perturbation approach, separation of variable, pressure head and flow force.

Chapter 1

Introduction

We consider a two dimensional, steady water waves in the channel of finite depth. We assume that the bottom of the channel is defined by $y = 0$ and $x \in R$. We describe the free surface by $y = \eta(x)$. Thus, the water domain W is defined by

$$W = \{(x, y) : x \in R,\ \ 0 < y < \eta(x)\},$$

where $\eta(x) > 0$ is a smooth function. We suppose that the density of the water is constant and equal to one. Since the flow is irrotational, the velocity field $(u,\, v)$ has a potential ϕ, i.e.

$$u = \phi_x, \quad \textit{and} \quad v = \phi_y$$

Since the velocity field satisfies

$$u_x + v_y = 0$$

the potential ϕ is a harmonic function. Let us introduce the conjugate function ψ connected with ϕ by the Cauchy Riemann equations:

$$u = -\psi_y, \quad \textit{and} \quad v = \psi_x. \tag{1.1}$$

Then ψ is also a harmonic function called the stream function. The normal derivative of velocity at the bottom and at the free surface is zero. This implies that at the bottom $v = 0$. Using (1.1), we obtain that $\psi_x = 0$ on the bottom. On the free surface, we have

$$un_1 + vn_2 = 0,$$

which implies

$$-\psi_y n_1 + \psi_x n_2 = 0,$$

$$\Leftrightarrow \bigtriangledown\psi.(n_2, -n_1) = 0,$$

therefore the tangent derivative of ψ is zero at the free surface

$$\frac{\partial\psi}{\partial\tau} = 0.$$

Thus ψ is constant on the bottom and on the free surface. Since ψ is defined up to a constant, we can assume that ψ is zero at the bottom, i.e.

$$\psi(x, 0) = 0,$$

and

$$\psi(x, \eta(x)) = Q,$$

where the quantity Q is constant, which is called the flux of the channel flow. The positive sign of Q shows that the waves propagate in the positive $x-$ direction. We assume that $Q = 1$. On the free surface, the velocity and η satisfy the Bernoulli's equation

$$\frac{1}{2}\left(u^2 + v^2\right) + g\eta(x) = \mathbb{R},$$

where g is the gravitational acceleration and we assume that it is equal to one and $\mathbb{R}$ is the Bernoulli constant for the flow. Using the definition of stream function, we arrive at

$$\frac{1}{2}|\bigtriangledown\psi(x, \eta(x))|^2 + \eta(x) = \mathbb{R}.$$

Thus the functions ψ and η satisfies the following boundary value problem

$$\begin{cases} \triangle\psi = 0, \quad \text{on} \quad W & \text{(1.2a)} \\ \psi(x, 0) = 0, \quad x \in [-\infty, \infty] & \text{(1.2b)} \\ \psi(x, \eta(x)) = 1, \quad x \in R & \text{(1.2c)} \\ \frac{1}{2}|\bigtriangledown\psi(x, \eta(x))|^2 + \eta(x) = \mathbb{R}, \quad x \in R. & \text{(1.2d)} \end{cases}$$

Our aim is to find ψ, η and $\mathbb{R}$ in the following form

$$\psi(x, y, \epsilon) = \psi_0(y) + \epsilon\psi_1(x, y) + \epsilon^2\psi_2(x, y) + O(\epsilon^3), \tag{1.3}$$

$$\eta(x, \epsilon) = \eta_0 + \epsilon\eta_1(x) + \epsilon^2\eta_2(x) + O(\epsilon^3), \tag{1.4}$$

$$\mathbb{R} = \mathbb{R}_0 + \epsilon\mathbb{R}_1 + \epsilon^2\mathbb{R}_2 + O(\epsilon^3), \tag{1.5}$$

where ϵ is a small parameter. In (1.3), we assume that ψ_0 depends only on y and in (1.4) we assume that η_0 is constant, denoted by h. In other words, (1.3-1.5) present the perturbation of a uniform stream. Our goal is to find the small periodic solution asymptotically. In the following figure(1.1), we

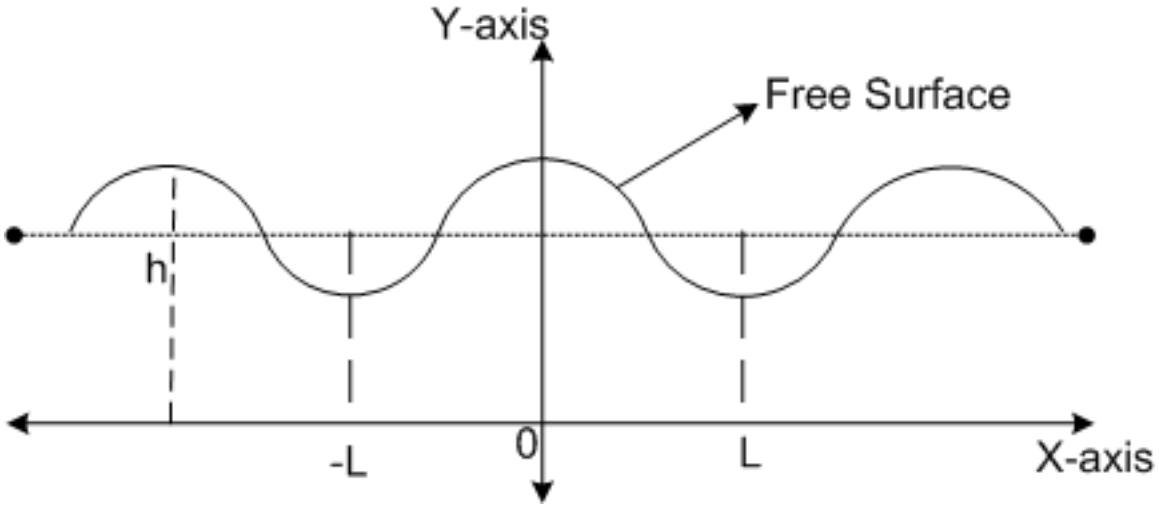


Figure 1.1: Periodic stoke's wave

consider steady periodic Stoke's wave with fixed period $2L$ and height h which is drawn in Cartesian coordinates. The dotted line represents the position of the free surface where no wave is present. The point of the wave which is the maximum of the free surface is called the crest and it is located at the origin. That point which is the minimum of the free surface is called the trough.

Chapter 2

Small periodic water waves with fixed period

2.1 Calculation of ψ_0, η_0 and $\mathbb{R}_0$

2.1.1 Asymptotic relations (1.2a)-(1.2d)

Using the fact that ψ is a harmonic and applying $\triangle$ to both sides of equation (1.3), we get

$$\triangle\psi(x,y,\epsilon) = \triangle\psi_0(y) + \epsilon \triangle \psi_1(x,y) + \epsilon^2 \triangle \psi_2(x,y) + O(\epsilon^3) = 0.$$

By equating the terms of order ϵ^0, ϵ^1 and ϵ^2, we have

$$\triangle\psi_0(y) = 0, \tag{2.1}$$

$$\triangle\psi_1(x,y) = 0, \tag{2.2}$$

$$\triangle\psi_2(x,y) = 0. \tag{2.3}$$

Putting $y = 0$ in equation (1.3) and taking into account (1.2b), we obtain

$$\psi(x,0,\epsilon) = \psi_0(0) + \epsilon\psi_1(x,0) + \epsilon^2\psi_2(x,0) + O(\epsilon^3) = 0,$$

from which

$$\psi_0(0) = 0, \tag{2.4}$$

$$\psi_1(x,0) = 0, \quad x \in R, \tag{2.5}$$

$$\psi_2(x,0) = 0, \quad x \in R. \tag{2.6}$$

Now (1.3) and (1.2c) give

$$\psi(x,\eta(x),\epsilon) = \psi_0(x,\eta(x)) + \epsilon\psi_1(x,\eta(x)) + \epsilon^2\psi_2(x,\eta(x)) + O(\epsilon^3) = 1.$$

Replacing $\eta(x)$ by its expression given in (1.4), we have

$$\psi_0(h+\epsilon\eta_1+\epsilon^2\eta_2)+\epsilon\psi_1(x,\eta_0+\epsilon\eta_1+\epsilon^2\eta_2)+\epsilon^2\psi_2(x,\eta_0+\epsilon\eta_1+\epsilon^2\eta_2)+O(\epsilon^3) = 1.$$

Applying the Taylor's expansion, we get

$$\psi_0(h) + \psi_{0y}(h)(\epsilon\eta_1 + \epsilon^2\eta_2) + \frac{1}{2}\psi_{0yy}(h)(\epsilon\eta_1 + \epsilon^2\eta_2)^2 + \epsilon\psi_1(x,h) + \epsilon\psi_{1y}(x,h)(\epsilon\eta_1 + \epsilon^2\eta_2) + \epsilon^2\psi_2(x,h) + O(\epsilon^3) = 1,$$

from which

$$\psi_0(h) = 1, \tag{2.7}$$

$$\psi_1(x,h) = -\psi_{0y}(h)\eta_1, \tag{2.8}$$

$$\psi_2(x,h) = -\psi_{0y}(h)\eta_2 - \frac{1}{2}\psi_{0yy}(h)\eta_1^2 - \psi_{1y}(x,h)\eta_1. \tag{2.9}$$

Using (1.2d), together with equations (1.3-1.5), we have

$$\frac{1}{2}\Big(\psi_{0x} + \epsilon\psi_{1x} + \epsilon^2\psi_{2x}\Big)^2 + \frac{1}{2}\Big(\psi_{0y} + \epsilon\psi_{1y} + \epsilon^2\psi_{2y}\Big)^2 + h + \epsilon\eta_1 + \epsilon^2\eta_1 = \mathbb{R}_0 + \epsilon\mathbb{R}_1 + \epsilon^2\mathbb{R}_2 + O(\epsilon^3).$$

Since we know that ψ_0 is a function of y only, its derivative with respect to x is zero. Therefore

$$\frac{1}{2}\Big(\epsilon\psi_{1x}(x,\eta(x)) + \epsilon^2\psi_{2x}(x,\eta(x))\Big)^2 + \frac{1}{2}\Big(\psi_{0y} + \epsilon\psi_{1y}(x,\eta(x)) + \epsilon^2\psi_{2y}(x,\eta(x))\Big)^2 + h + \epsilon\eta_1 + \epsilon^2\eta_1 = \mathbb{R}_0 + \epsilon\mathbb{R}_1 + \epsilon^2\mathbb{R}_2 + O(\epsilon^3),$$

expanding the squares, we get

$$\frac{1}{2}\epsilon^2\psi_{1x}^2(x,\eta(x)) + \frac{1}{2}\Big((\psi_{0y})^2 + \epsilon^2\psi_{1y}^2(x,\eta(x)) + 2\epsilon\psi_{0y}\psi_{1y}(x,\eta(x)) + 2\psi_{0y}\epsilon^2\psi_{2y}(x,\eta(x))\Big) + h + \epsilon\eta_1 + \epsilon^2\eta_1 = \mathbb{R}_0 + \epsilon\mathbb{R}_1 + \epsilon^2\mathbb{R}_2 + O(\epsilon^3).$$

Using (1.4) in the above expression, we have

$$\frac{1}{2}\epsilon^2\psi_{1x}^2(x,h+\epsilon\eta_1+\epsilon^2\eta_2)+\frac{1}{2}\bigg((\psi_{0y})^2+\epsilon^2\psi_{1y}^2(x,h+\epsilon\eta_1+\epsilon^2\eta_2)+$$
$$2\epsilon\psi_{0y}\psi_{1y}(x,h+\epsilon\eta_1+\epsilon^2\eta_2)+2\epsilon^2\psi_{0y}\psi_{2y}(x,h+\epsilon\eta_1+\epsilon^2\eta_2)\bigg)+h+$$
$$\epsilon\eta_1(x)+\epsilon^2\eta_2(x)=\mathbb{R}_0+\epsilon\mathbb{R}_1+\epsilon^2\mathbb{R}_2+O(\epsilon^3).$$

Applying Taylor's expansion, we get

$$\frac{1}{2}\epsilon^2\bigg(\psi_{1x}(x,h)+\psi_{1xy}(x,h)(\epsilon\eta_1+\epsilon^2\eta_2)\bigg)^2+\frac{1}{2}\bigg((\psi_{0y})^2+\epsilon^2\bigg(\psi_{1y}(x,h)+$$
$$\psi_{1yy}(x,h)(\epsilon\eta_1+\epsilon^2\eta_2)\bigg)^2+2\epsilon\psi_{0y}\bigg(\psi_{1y}(x,h)+\psi_{1yy}(x,h)(\epsilon\eta_1+\epsilon^2\eta_2)\bigg)+$$
$$2\epsilon^2\psi_{0y}\bigg(\psi_{2y}(x,h)+\psi_{2yy}(x,h)(\epsilon\eta_1+\epsilon^2\eta_2)\bigg)\bigg)+h+\epsilon\eta_1(x)+\epsilon^2\eta_2(x)=$$
$$\mathbb{R}_0+\epsilon\mathbb{R}_1+\epsilon^2\mathbb{R}_2+O(\epsilon^3),$$

from which

$$\mathbb{R}_0=\frac{1}{2}\psi_{0y}^2(h)+h, \tag{2.10}$$

$$\mathbb{R}_1=\psi_{0y}\psi_{1y}(x,h)+\eta_1(x), \tag{2.11}$$

$$\mathbb{R}_2=\frac{1}{2}\psi_{1x}^2(x,h)+\frac{1}{2}\psi_{1y}^2(x,h)+\psi_{0y}\psi_{1yy}(x,h)\eta_1+\psi_{oy}\psi_{2y}(x,h)+\eta_2. \tag{2.12}$$

2.1.2 Boundary value problem for ψ_0, η_0 and $\mathbb{R}_0$

Now combining equations (2.1), (2.4), (2.7) and (2.10), we formulate the following boundary value problem

$$\begin{cases} \triangle\psi_0(y)=0, & \text{(2.13a)}\\ \psi_0(0)=0, \quad y\in[0,h] & \text{(2.13b)}\\ \psi_0(h)=1, \quad y\in[0,h] & \text{(2.13c)}\\ \frac{1}{2}\bigg(\psi_{0y}(h)\bigg)^2+h=\mathbb{R}_0. & \text{(2.13d)} \end{cases}$$

2.1.3 Solution to (2.13)

Solving (2.13) for ψ_0 with first two boundary conditions, we obtain

$$\psi_0(y) = \frac{y}{h}. \tag{2.14}$$

Now substituting the obtained ψ_0 in (2.13d), we get

$$\frac{1}{2h^2} + h = \mathbb{R}_0. \tag{2.15}$$

Let us consider a function $R(h)$ defined by

$$\frac{1}{2h^2} + h = R(h),$$

see in graph at figure (2.1). We observe that $R(h)$ has its minimum at $h = 1$. Therefore equation (2.15) has solutions only when $\mathbb{R}_0 \geq \frac{3}{2}$. We assume that $\mathbb{R}_0 > \frac{3}{2}$, under this condition, the equation (2.15) has two solutions. If we choose one of them, then the function ψ_0 given by (2.14), $\eta_0 = h$ and $\mathbb{R}_0$ solve problem (2.13a-2.13d).

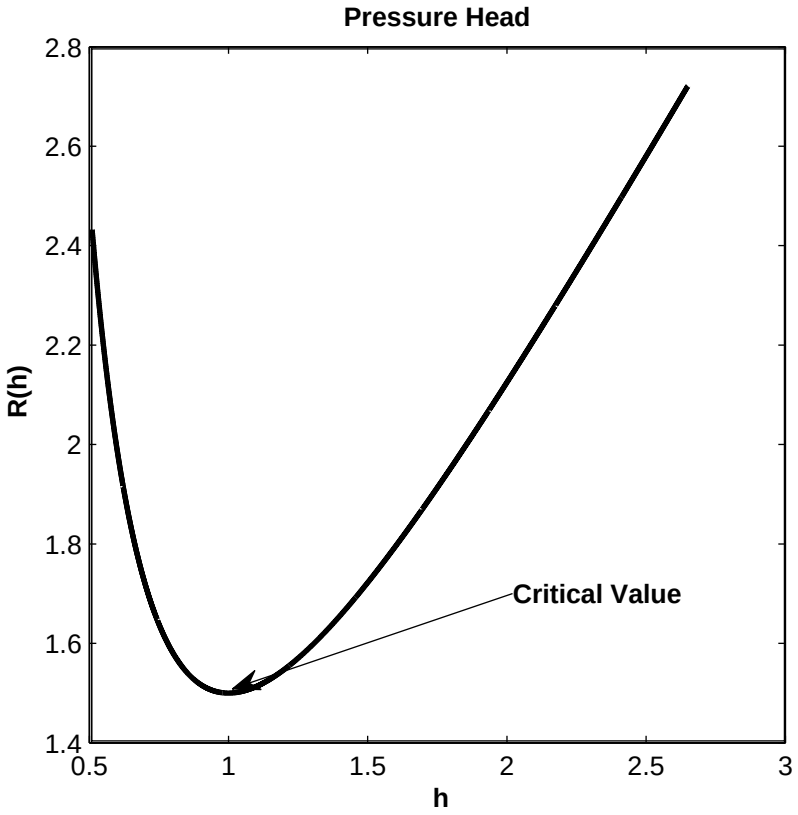


Figure 2.1: Overview of R(h)

2.2 Calculation of ψ_1, η_1 and $\mathbb{R}_1$

2.2.1 Boundary value problem for ψ_1, η_1 and $\mathbb{R}_1$

Using (2.14) in the equation (2.8), we get

$$\psi_1(x,h) = -\frac{\eta_1(x)}{h}. \tag{2.16}$$

Using (2.14) in (2.11), we have

$$\mathbb{R}_1 = \frac{1}{h}\psi_{1y}(x,h) + \eta_1(x). \tag{2.17}$$

Now combining equations (2.2), (2.5), (2.16) and (2.17), we formulate the following boundary value problem

$$\begin{cases} \triangle\psi_1(x,y) = 0, & \text{(2.18a)} \\ \psi_1(x,0) = 0, \quad x \in [-L,L] & \text{(2.18b)} \\ \psi_1(x,h) = -\dfrac{\eta_1}{h},\ x \in [-L,L] & \text{(2.18c)} \\ \dfrac{1}{h}\psi_{1y}(x,h) + \eta_1 = \mathbb{R}_1, \quad x \in [-L,L]. & \text{(2.18d)} \end{cases}$$

2.2.2 Solution to (2.18)

We seek a solution to (2.18) in the following form

$$\psi_1(x,y) = \gamma(y)\cos(\frac{x\pi}{L}), \tag{2.19}$$

$$\eta_1(x) = \cos(\frac{x\pi}{L}), \tag{2.20}$$

where $\gamma(y)$ is unknown function which we need to find and L is also unknown with half period. Applying $\triangle$ to both sides of (2.19) and using the fact that ψ_1 is harmonic, we have

$$\triangle\psi_1(x,y) = \gamma''(y) - \frac{\pi^2}{L^2}\gamma(y) = 0. \tag{2.21}$$

Making use of (2.19) in (2.18b), we have

$$\psi_1(x,0) = \gamma(0)\cos(\frac{x\pi}{L}) = 0,$$

from which

$$\gamma(0) = 0. \tag{2.22}$$

Making use of (2.19) in (2.18c), we have

$$\psi_1(x,h) = \gamma(h)\cos(\frac{x\pi}{L}) = -\frac{\eta_1}{h}.$$

Replacing η_1 by its expression given in (2.20), we have

$$\gamma(h) = -\frac{1}{h}. \tag{2.23}$$

Now combining equations (2.21)-(2.23), we have the following ordinary differential equation

$$\begin{cases} \gamma''(y) - \dfrac{\pi^2}{L^2}\gamma(y) = 0, \\ \\ \gamma(0) = 0, \quad y = 0 \\ \\ \gamma(h) = -\frac{1}{h}, \quad y = h. \end{cases} \tag{2.24}$$

Solving the above system of equations, we get

$$\gamma(y) = \frac{-1}{h}\frac{\sinh(\frac{y\pi}{L})}{\sinh(\frac{h\pi}{L})}.$$

Substituting $\gamma(y)$ in (2.19), we obtain

$$\psi_1(x,y) = -\frac{1}{h}\frac{\sinh(\frac{\pi y}{L})}{\sinh(\frac{\pi h}{L})}\cos(\frac{x\pi}{L}), \tag{2.25}$$

therefore from (2.18d)

$$\mathbb{R}_1 = \frac{1}{h}\left(\frac{-\pi}{Lh}\cos(\frac{x\pi}{L})\frac{1}{\tanh(\frac{h\pi}{L})}\right) + \cos(\frac{x\pi}{L}),$$

or

$$\mathbb{R}_1 = \left(\frac{-\pi}{h^2 L}\frac{1}{\tanh(\frac{h\pi}{L})} + 1\right)\cos(\frac{x\pi}{L}). \tag{2.26}$$

This equation can be satisfied only when

$$\frac{-\pi}{h^2 L}\frac{1}{\tanh(\frac{h\pi}{L})} + 1 = 0, \tag{2.27}$$

and hence $\mathbb{R}_1 = 0$. Relation (2.27) is called the dispersion relation and the half period L satisfies it. Putting $\tau = \dfrac{h\pi}{L}$, in equation (2.27), we have

$$\frac{\tanh\tau}{\tau} = \frac{1}{h^3}. \tag{2.28}$$

Since

$$\frac{\tanh\tau}{\tau} < 1,$$

then

$$\frac{1}{h^3} < 1,$$

which implies that h should be greater than 1. Thus, in order to satisfy (2.27) we must choose the root of equation (2.15) which is greater than one. Therefore $h > 1$ and the equation (2.28) has a solution τ and then L is given by

$$L = \frac{h\pi}{\tau}. \tag{2.29}$$

Thus we have found η_1, ψ_1, $\mathbb{R}_1$ and L which are given by equations (2.20), (2.25), (2.26) and (2.29) respectively.

2.3 Calculation of ψ_2, η_2 and $\mathbb{R}_2$

2.3.1 Boundary value problem for ψ_2, η_2 and $\mathbb{R}_2$

Using (2.14) and (2.25) in equation (2.9), we get

$$\psi_2(x,h) = -\frac{\eta_2}{h} + \frac{h}{2} + \frac{h}{2}\cos(\frac{2\pi x}{L}). \tag{2.30}$$

Using (2.14) and (2.25) in equation (2.12), we obtain

$$\frac{\pi^2}{2h^2L^2}\sin^2(\frac{\pi x}{L}) + \frac{h^2}{2}\cos^2(\frac{\pi x}{L}) - \frac{\pi^2}{h^2L^2}\cos^2(\frac{\pi x}{L}) + \frac{1}{h}\psi_{2y}(x,h) + \eta_2 = \mathbb{R}_2. \tag{2.31}$$

Since

$$\sin^2(\frac{\pi x}{L}) = \frac{1}{2}\bigg(1 - cos(\frac{2\pi x}{L})\bigg),$$

using above relation in equation (2.31), we have

$$\frac{\pi^2}{4h^2L^2}\bigg(1-\cos(\frac{2\pi x}{L})\bigg)+\bigg(\frac{h^2}{4}-\frac{\pi^2}{2h^2L^2}\bigg)\bigg(1+\cos(\frac{2\pi x}{L})\bigg)+\frac{1}{h}\psi_{2y}(x,h)+\eta_2 = \mathbb{R}_2. \tag{2.32}$$

Now combining equations (2.3), (2.6), (2.30) and (2.32), we formulate the following boundary value problem

$$\begin{cases} \triangle\psi_2(x,y) = 0, \\ \\ \psi_2(x,0) = 0, \quad x \in [-L, L] \\ \\ \psi_2(x,h) = -\dfrac{\eta_2}{h} + \dfrac{h}{2} + \dfrac{h}{2} cos(\frac{2\pi x}{L}), \quad x \in [-L, L] \\ \\ \dfrac{\pi^2}{4h^2L^2}\left(1 - \cos(\frac{2\pi x}{L})\right) + \left(\dfrac{h^2}{4} - \dfrac{\pi^2}{2h^2L^2}\right)\left(1 + \cos(\frac{2\pi x}{L})\right) \\ \\ +\dfrac{1}{h}\psi_{2y}(x,h) + \eta_2 = \mathbb{R}_2. \end{cases} \tag{2.33}$$

2.3.2 Solution to (2.33)

We seek a solution $\psi_2(x,y)$ in the following form

$$\psi_2(x,y) = \psi^{(1)}(y) + \psi^{(2)} \cos(\frac{2\pi x}{L}), \tag{2.34}$$

$$\eta_2(x) = \eta^{(1)} + \eta^{(2)} \cos(\frac{2\pi x}{L}). \tag{2.35}$$

Applying $\triangle$ to both sides of equation (2.34) and using the fact that ψ_2 is harmonic, we have

$$\triangle\psi_2(x,h) = \psi^{(1)}_{yy} + \left(\psi^{(2)}_{yy} - \psi^{(2)}(\frac{4\pi^2}{L^2})\right)\cos(\frac{2\pi x}{L}) = 0,$$

from which

$$\psi^{(2)}_{yy} - (\frac{4\pi^2}{L^2})\psi^{(2)} = 0, \tag{2.36}$$

and

$$\psi^{(1)}_{yy} = 0. \tag{2.37}$$

Making use of equation (2.34) in 2nd equation of (2.33), we have

$$\psi_2(x,0) = \psi^{(1)}(0) + \psi^{(2)} \cos(\frac{2\pi x}{L}) = 0,$$

from which

$$\psi^{(1)}(0) = 0, \tag{2.38}$$

and

$$\psi^{(2)}(0) = 0. \tag{2.39}$$

Making use of equation (2.34) in 3rd equation of (2.33), we have

$$\psi_2(x,h) = \psi^{(1)}(h) + \psi^{(2)}\cos(\frac{2\pi x}{L}) = -\frac{\eta_2}{h} + \frac{h}{2} + \frac{h}{2}cos(\frac{2\pi x}{L}).$$

Replacing η_2 by its expression given in (2.35), we have

$$\psi^{(1)}(h) = -\frac{\eta^{(1)}}{h} + \frac{h}{2}, \tag{2.40}$$

and

$$\psi^{(2)}(h) = -\frac{\eta^{(2)}}{h} + \frac{h}{2}. \tag{2.41}$$

Making use of equations (2.34) and (2.35) in 4th equation of (2.33), we have

$$\frac{\pi^2}{4h^2L^2}\left(1-\cos(\frac{2\pi x}{L})\right) + \left(\frac{h^2}{4} - \frac{\pi^2}{2h^2L^2}\right)\left(1+\cos(\frac{2\pi x}{L})\right) + \frac{1}{h}\left(\psi_y^{(1)} + \psi_y^{(2)}\cos(\frac{2\pi x}{L})\right) + \left(\eta^{(1)} + \eta^{(2)}\cos(\frac{2\pi x}{L})\right) = \mathbb{R}_2,$$

from which

$$\frac{\pi^2}{4h^2L^2} + (\frac{h^2}{4} - \frac{\pi^2}{2h^2L^2}) + \frac{1}{h}\psi_y^{(1)} + \eta^{(1)} = \mathbb{R}_2, \tag{2.42}$$

and

$$-\frac{\pi^2}{4h^2L^2} + (\frac{h^2}{4} - \frac{\pi^2}{2h^2L^2}) + \frac{1}{h}\psi_y^{(2)} + \eta^{(2)} = 0. \tag{2.43}$$

Now combining equations (2.37), (2.38), (2.40) and (2.42), we formulate the following ordinary differential equation

$$\begin{cases} \psi_{yy}^{(1)} = 0, \\ \psi^{(1)}(0) = 0, \\ \psi^{(1)}(h) = -\dfrac{\eta^{(1)}}{h} + \dfrac{h}{2}, \\ \dfrac{\pi^2}{4h^2L^2} + \left((\dfrac{h^2}{4} - \dfrac{\pi^2}{2h^2L^2}) + \dfrac{1}{h}\psi_y^{(1)}(x,h)\right) + \eta^{(1)} = \mathbb{R}_2. \end{cases} \tag{2.44}$$

We assume that $\eta^{(1)}$ is zero. Solving (2.44) for $\psi^{(1)}$ with first two boundary conditions, we obtain

$$\psi^{(1)}(y) = -\frac{y}{2}.$$

Substituting above value in forth equation of (2.44), we get

$$-\frac{\pi^2}{4h^2L^2} + \frac{h^2}{4} + \frac{1}{h}\frac{1}{2} = \mathbb{R}_2,$$

or

$$\mathbb{R}_2 = \frac{h^2}{4}\left(1 - \tanh^2(\frac{\pi h}{L})\right) + \frac{1}{2h}. \tag{2.45}$$

Now combining equations (2.36), (2.39), (2.41) and (2.43), we formulate the following ordinary differential equation

$$\begin{cases} \psi^{(2)}_{yy} - \dfrac{4\pi^2}{L^2}\psi^{(2)} = 0, \\ \\ \psi^{(2)}(0) = 0, \\ \\ \psi^{(2)}(h) = -\dfrac{\eta^{(2)}}{h} + \dfrac{h}{2}, \\ \\ -\dfrac{\pi^2}{4h^2L^2} + (\dfrac{h^2}{4} - \dfrac{\pi^2}{2h^2L^2}) + \dfrac{1}{h}\psi^{(2)}_y(x,h) + \eta^{(2)} = 0. \end{cases} \tag{2.46}$$

Solving (2.46) for $\psi^{(2)}$ with first two boundary conditions, we obtain

$$\psi^{(2)} = (-\frac{\eta^{(2)}}{h} + \frac{h}{2})\frac{1}{2}\frac{(e^{\frac{2\pi y}{L}} - e^{\frac{-2\pi y}{L}})}{\sinh(\frac{2\pi h}{L})},$$

after some simplification,

$$\psi^{(2)} = (-\frac{\eta^{(2)}}{h} + \frac{h}{2})\frac{\sinh(\frac{2\pi y}{L})}{\sinh(\frac{2\pi h}{L})}. \tag{2.47}$$

Using (2.47) in forth equation of (2.46), we have

$$-\frac{\pi^2}{4h^2L^2} + (\frac{h^2}{4} - \frac{\pi^2}{2h^2L^2}) + \frac{1}{h}(-\frac{\eta^{(2)}}{h} + \frac{h}{2})\frac{2\pi}{L}\frac{\cos h(\frac{2\pi h}{L})}{\sinh(\frac{2\pi h}{L})} + \eta^{(2)} = 0,$$

or

$$-\frac{\pi^2}{4h^2L^2}+(\frac{h^2}{4}-\frac{\pi^2}{2h^2L^2})+\frac{2\pi}{hL}\left(-\frac{\eta^{(2)}}{h}+\frac{h}{2}\right)\coth(\frac{2\pi h}{L})+\eta^{(2)}=0. \quad (2.48)$$

Since

$$\tanh(\frac{2\pi h}{L})=\frac{2tanh(\frac{\pi h}{L})}{1+tanh^2(\frac{\pi h}{L})},$$

and also from (2.27), we have equation (2.48) in the following form

$$\frac{-3\pi^2}{4L^2h^2}+\frac{h^2}{4}+\left(-\frac{\eta^{(2)}}{h}+\frac{h}{2}\right)h(1+\frac{\pi^2}{L^2h^4})+\eta^{(2)}=0,$$

or

$$\frac{-3\pi^2}{4L^2h^2}+\frac{h^2}{4}+\eta^{(2)}\left(1-1-\frac{\pi^2}{L^2h^4}\right)+\frac{h^2}{2}\left(1+\frac{\pi^2}{L^2h^4}\right)=0,$$

therefore

$$\eta^{(2)}=\frac{h^2}{4}\left(\frac{3L^2h^4}{\pi^2}-1\right). \quad (2.49)$$

Substituting $\eta^{(1)}$ and $\eta^{(2)}$ in (2.35), we get

$$\eta_2(x)=\frac{h^2}{4}\left(\frac{3L^2h^4}{\pi^2}-1\right)\cos(\frac{2\pi x}{L}). \quad (2.50)$$

Now substituting $\psi^{(1)}$ and $\psi^{(2)}$ in equation (2.34), we get

$$\psi_2(x,y)=\frac{y}{2}+\left(\frac{h}{2}-\frac{h}{4}(\frac{3h^4L^2}{\pi^2}-1)\right)\frac{\sinh\frac{2\pi y}{L}}{\sinh(\frac{2\pi h}{L})}\cos(\frac{2\pi x}{L}),$$

or

$$\psi_2(x,y)=\frac{3h}{4}\left(1-\frac{h^4L^2}{\pi^2}\right)\frac{\sinh(\frac{2\pi y}{L})}{\sinh(\frac{2\pi h}{L})}\cos(\frac{2\pi x}{L})+\frac{y}{2}. \quad (2.51)$$

Thus we have found $\mathbb{R}_2$, $\eta_2(x)$ and $\psi_2(x,y)$ which are given by equations (2.45), (2.50) and (2.51) respectively. We compare our results with [2] which are mentioned in equations (2.45), (2.50) and (2.51) and we get the required results.

Chapter 3

Small periodic water waves with fixed Bernoulli constant

Our goal is to find the small amplitude solution with fixed Bernoulli constant $\mathbb{R}$, but the free surface, stream function and the period are unknown. In this case it is more convenient to look at the solution to (1.2) in the following form

$$\psi(\mathbb{X}, y, \epsilon) = \psi_0(y) + \epsilon\psi_1(\mathbb{X}, y) + \epsilon^2\psi_2(\mathbb{X}, y) + O(\epsilon^3), \tag{3.1}$$

$$\eta(\mathbb{X}, \epsilon) = \eta_0 + \epsilon\eta_1(\mathbb{X}) + \epsilon^2\eta_2(\mathbb{X}) + O(\epsilon^3), \tag{3.2}$$

$$\mathbb{L} = \mathbb{L}_0 + \epsilon\mathbb{L}_1 + \epsilon^2\mathbb{L}_2 + O(\epsilon^3), \tag{3.3}$$

where $\mathbb{X} = \dfrac{x\mathbb{L}_0}{\mathbb{L}}$. One can check that if

$$x \in [-\mathbb{L}(\epsilon), \mathbb{L}(\epsilon)]$$

then

$$\mathbb{X} \in [-\mathbb{L}_0, \mathbb{L}_0].$$

The boundary value problem (1.2) can be written in new variable as

$$\begin{cases} (\dfrac{\mathbb{L}_0}{\mathbb{L}})^2\psi_{\mathbb{X}\mathbb{X}} + \psi_{yy} = 0, \quad \text{on} \quad W & \text{(3.4a)} \\ \psi(\dfrac{\mathbb{X}\mathbb{L}}{\mathbb{L}_0}, 0) = 0, \ \ \mathbb{X} \in [-\mathbb{L}_0, \mathbb{L}_0] & \text{(3.4b)} \\ \psi(\dfrac{\mathbb{X}\mathbb{L}}{\mathbb{L}_0}, \eta(\dfrac{\mathbb{X}\mathbb{L}}{\mathbb{L}_0})) = 1, & \text{(3.4c)} \\ \dfrac{1}{2}|(\dfrac{\mathbb{L}_0}{\mathbb{L}})\psi_{\mathbb{X}} + \psi_y|^2 + \eta(\mathbb{X}) = \mathbb{R} \ \ \mathbb{X} \in [-\mathbb{L}_0, \mathbb{L}_0]. & \text{(3.4d)} \end{cases}$$

Our aim is to find ψ, η and $\mathbb{R}$ asymptotically.

3.1 Calculation of ψ_0, η_0 and $\mathbb{R}$

3.1.1 Asymptotic relations (3.4a)-(3.4d)

Using (3.1) in (3.4a), we get

$$\psi_{0yy}(y)+\epsilon\bigg((\frac{\mathbb{L}_0}{\mathbb{L}})^2\psi_{1\mathbb{X}\mathbb{X}}(\mathbb{X},y)+\psi_{1yy}\bigg)+\epsilon^2\bigg((\frac{\mathbb{L}_0}{\mathbb{L}})^2\psi_{2\mathbb{X}\mathbb{X}}(\mathbb{X},y)+\psi_{2yy}\bigg)+ O(\epsilon^3)=0.$$

Using (3.3), we obtain

$$\psi_{0yy}+\epsilon\bigg((1+\epsilon\frac{\mathbb{L}_1}{\mathbb{L}_0}+\epsilon^2\frac{\mathbb{L}_2}{\mathbb{L}_0})^{-2}\psi_{1\mathbb{X}\mathbb{X}}+\psi_{1yy}\bigg)+\epsilon^2\bigg((1+\epsilon\frac{\mathbb{L}_1}{\mathbb{L}_0}+\epsilon^2\frac{\mathbb{L}_2}{\mathbb{L}_0})^{-2}\psi_{2\mathbb{X}\mathbb{X}} +\psi_{2yy}\bigg)+O(\epsilon^3)=0,$$

or

$$\psi_{0yy}+\epsilon\bigg((1-2(\epsilon\frac{\mathbb{L}_1}{\mathbb{L}_0}+\epsilon^2\frac{\mathbb{L}_2}{\mathbb{L}_0}))\psi_{1\mathbb{X}\mathbb{X}}+\psi_{1yy}\bigg)+\epsilon^2\bigg((1-2(\epsilon\frac{\mathbb{L}_1}{\mathbb{L}_0}+\epsilon^2\frac{\mathbb{L}_2}{\mathbb{L}_0}))\psi_{2\mathbb{X}\mathbb{X}} +\psi_{2yy}\bigg)+O(\epsilon^3)=0,$$

therefore

$$\psi_{0yy}+\epsilon\bigg(\psi_{1\mathbb{X}\mathbb{X}}-2(\epsilon\frac{\mathbb{L}_1}{\mathbb{L}_0}+\epsilon^2\frac{\mathbb{L}_2}{\mathbb{L}_0})\psi_{1\mathbb{X}\mathbb{X}}+\psi_{1yy}\bigg)+\epsilon^2\bigg(\psi_{2\mathbb{X}\mathbb{X}}-2(\epsilon\frac{\mathbb{L}_1}{\mathbb{L}_0}+\epsilon^2\frac{\mathbb{L}_2}{\mathbb{L}_0})\psi_{2\mathbb{X}\mathbb{X}} +\psi_{2yy}\bigg)+O(\epsilon^3)=0.$$

By equating the terms of order ϵ^0, ϵ^1 and ϵ^2, we have

$$\psi_{0yy}=0, \tag{3.5}$$

$$\psi_{1\mathbb{X}\mathbb{X}}+\psi_{1yy}=0, \tag{3.6}$$

$$\frac{-2\mathbb{L}_1}{\mathbb{L}_0}\psi_{1\mathbb{X}\mathbb{X}}+\psi_{2\mathbb{X}\mathbb{X}}+\psi_{2yy}=0. \tag{3.7}$$

Putting $y=0$ in equation (3.1) and taking into account (3.4b), we obtain

$$\psi(\mathbb{X},0,\epsilon)=\psi_0(0)+\epsilon\psi_1(\mathbb{X},0)+\epsilon^2\psi_2(\mathbb{X},0)+O(\epsilon^3)=0,$$

from which

$$\psi_0(0) = 0, \tag{3.8}$$

$$\psi_1(\mathbb{X}, 0) = 0, \tag{3.9}$$

$$\psi_2(\mathbb{X}, 0) = 0. \tag{3.10}$$

Using(3.1-3.3) in (3.4c), we have

$$\psi_0(h + \epsilon\eta_1 + \epsilon^2\eta_2) + \epsilon\psi_1(\mathbb{X}, h + \epsilon\eta_1 + \epsilon^2\eta_2) + \epsilon^2\psi_2(\mathbb{X}, h + \epsilon\eta_1 + \epsilon^2\eta_2) + O(\epsilon^3) = 1.$$

Applying Taylor's expansion, we get

$$\psi_0(h) + \psi_{0y}(\epsilon\eta_1 + \epsilon^2\eta_2) + \frac{1}{2}\psi_{0yy}(\epsilon\eta_1 + \epsilon^2\eta_2)^2 + \epsilon\Big(\psi_1(\mathbb{X}, h) + \psi_{1y}(\mathbb{X}, h)(\epsilon\eta_1 + \epsilon^2\eta_2)\Big) + \epsilon^2\psi_2(\mathbb{X}, h) + O(\epsilon^3) = 1,$$

from which

$$\psi_0(h) = 1, \tag{3.11}$$

$$\psi_{0y}(h)\eta_1 + \psi_1(\mathbb{X}, h) = 0, \tag{3.12}$$

$$\psi_{0y}\eta_2 + \frac{1}{2}\psi_{0yy}\eta_1^2 + \psi_{1y}(\mathbb{X}, h)\eta_2 + \psi_2(\mathbb{X}, h) = 0. \tag{3.13}$$

Using (3.4d), together with (3.1-3.3), we have

$$\frac{1}{2}\left(\frac{\mathbb{L}_0}{\mathbb{L}}\right)^2\Big(\psi_{0\mathbb{X}} + \epsilon\psi_{1\mathbb{X}} + \epsilon^2\psi_{2\mathbb{X}}\Big)^2 + \frac{1}{2}\Big(\psi_{0y} + \epsilon\psi_{1y} + \epsilon^2\psi_{2y}\Big)^2 + h + \epsilon\eta_1 + \epsilon^2\eta_1 + O(\epsilon^3) = \mathbb{R}.$$

Since we know that ψ_0 is a function of y only, its derivative with respect to x is zero,

$$\frac{1}{2}\Big(1 - 2(\epsilon\frac{\mathbb{L}_1}{\mathbb{L}_0} + \epsilon^2\frac{\mathbb{L}_2}{\mathbb{L}_0})\Big)\epsilon^2\psi_{1\mathbb{X}}^2 + \frac{1}{2}\Big(\psi_{0y} + \epsilon\psi_{1y} + \epsilon^2\psi_{2y}\Big)^2 + h + \epsilon\eta_1 + \epsilon^2\eta_1 + O(\epsilon^3) = \mathbb{R}.$$

Applying Taylor's expansion, we have

$$\left(\frac{\epsilon^2}{2}\psi_{1\mathbb{X}}^2 - \epsilon^2\psi_{1\mathbb{X}}^2(\epsilon\frac{\mathbb{L}_1}{\mathbb{L}_0} + \epsilon^2\frac{\mathbb{L}_2}{\mathbb{L}_0})\right) + \frac{\psi_{0y}^2}{2} + \frac{\epsilon^2}{2}\left(\psi_{1y}(\mathbb{X},h) + \psi_{1yy}(\mathbb{X},h)(\epsilon\eta_1 + \epsilon^2\eta_2)\right)^2$$
$$+\frac{\epsilon}{h}\left(\psi_{1y}(\mathbb{X},h) + \psi_{1yy}(\mathbb{X},h)(\epsilon\eta_1 + \epsilon^2\eta_2)\right) + \frac{\epsilon^2}{h}\left(\psi_{2y}(\mathbb{X},h) + \psi_{2yy}(\mathbb{X},h)(\epsilon\eta_1 + \epsilon^2\eta_2)\right)$$
$$+h + \epsilon\eta_1 + \epsilon^2\eta_2 + O(\epsilon^3) = \mathbb{R},$$

from which

$$\frac{\psi_{0y}^2}{2} + h = \mathbb{R}, \tag{3.14}$$

$$\frac{1}{h}\psi_{1y}(\mathbb{X},h) + \eta_1 = 0, \tag{3.15}$$

$$\frac{1}{2}\psi_{1\mathbb{X}}^2(\mathbb{X},h) + \frac{1}{2}\left(\psi_{1y}^2(\mathbb{X},h) + \frac{2}{h}\psi_{1yy}(\mathbb{X},h) + \frac{2}{h}\psi_{2y}(\mathbb{X},h)\right) + \eta_2 = 0. \tag{3.16}$$

3.1.2 Boundary value problem for ψ_0, η_0 and $\mathbb{R}$

Now combining equations (3.5), (3.8), (3.11) and (3.14), we formulate the following boundary value problem

$$\begin{cases} \psi_{0yy}(y) = 0, & \text{(3.17a)} \\ \psi_0(0) = 0, \quad \mathbb{X} \in [-\mathbb{L}_0, \mathbb{L}_0] & \text{(3.17b)} \\ \psi_0(h) = 1, & \text{(3.17c)} \\ \dfrac{\psi_{0y}^2}{2} + h = \mathbb{R}, \quad \mathbb{X} \in [-\mathbb{L}_0, \mathbb{L}_0]. & \text{(3.17d)} \end{cases}$$

3.1.3 Solution to (3.17)

Solving (3.17a) for ψ_0 with first two boundary conditions, we obtain

$$\psi_0(y) = \frac{y}{h}. \tag{3.18}$$

Now substituting the obtained ψ_0 in (3.17d), we get

$$\frac{1}{2h^2} + h = \mathbb{R}, \tag{3.19}$$

see in graph at figure (2.1). $\mathbb{R}$ has minimum at $h = 1$, therefore (3.19) has solutions only when $\mathbb{R} \geq \frac{3}{2}$. We assume that $\mathbb{R} > \frac{3}{2}$, under this condition,

the equation (3.19) has two solutions. If we choose one of them, then the function ψ_0 given by (3.18), $\eta_0 = h$ and $\mathbb{R}$ solve the problem (3.17a)-(3.17d). Thus we have found ψ_0 and $\mathbb{R}$ which are given by equations (3.18) and (3.19) respectively.

3.2 Calculation of ψ_1, η_1 and $\mathbb{L}_0$

3.2.1 Boundary value problem for ψ_1, η_1 and $\mathbb{L}_0$

Using (3.18) in (3.12), we get

$$\psi_1(\mathbb{X}, h) = -\frac{\eta_1(\mathbb{X})}{h}. \tag{3.20}$$

Now combining equations (3.6), (3.9), (3.15) and (3.20), we formulate the following boundary value problem

$$\begin{cases} \psi_{1\mathbb{X}\mathbb{X}}(\mathbb{X}, y) + \psi_{1yy}(\mathbb{X}, y) = 0, & \text{(3.21a)} \\ \psi_1(\mathbb{X}, 0) = 0, \quad \mathbb{X} \in [\mathbb{L}_0, \mathbb{L}_0] & \text{(3.21b)} \\ \psi_1(\mathbb{X}, h) = -\dfrac{\eta_1}{h}, & \text{(3.21c)} \\ \dfrac{1}{h}\psi_{1y}(\mathbb{X}, h) + \eta_1 = 0, \quad \mathbb{X} \in [\mathbb{L}_0, \mathbb{L}_0]. & \text{(3.21d)} \end{cases}$$

3.2.2 Solution to (3.21)

We seek a solution to (3.21) in the following form

$$\psi_1(\mathbb{X}, y) = \gamma(y)\cos(\frac{\mathbb{X}\pi}{\mathbb{L}_0}), \tag{3.22}$$

$$\eta_1(\mathbb{X}) = \cos(\frac{\mathbb{X}\pi}{\mathbb{L}_0}), \tag{3.23}$$

where $\gamma(y)$ and $\mathbb{L}_0$ are unknown functions which we need to find. Using (3.22) in (3.21a), we have

$$\psi_1(\mathbb{X}, y) = \gamma''(y) - \frac{\pi^2}{\mathbb{L}_0^2}\gamma(y) = 0. \tag{3.24}$$

Using (3.22) in (3.21b), we get

$$\psi_1(\mathbb{X}, 0) = \gamma(0)\cos(\frac{\mathbb{X}\pi}{\mathbb{L}_0}) = 0, \tag{3.25}$$

from which

$$\gamma(0) = 0. \tag{3.26}$$

Using (3.22) in (3.21c), we have

$$\psi_1(\mathbb{X}, h) = \gamma(h)\cos(\frac{\mathbb{X}\pi}{\mathbb{L}_0}) = -\frac{\eta_1(\mathbb{X})}{h}, \tag{3.27}$$

Replacing η_1 by its expression given in (3.23), we have

$$\gamma(h) = -\frac{1}{h} \tag{3.28}$$

Now combining equations (3.24), (3.26) and (3.28), we have the following ordinary differential equation

$$\begin{cases} \gamma''(y) - \dfrac{\pi^2}{\mathbb{L}_0^2}\gamma(y) = 0 & \text{(3.29a)} \\ \gamma(0) = 0, \quad y = 0 & \text{(3.29b)} \\ \gamma(h) = -\dfrac{1}{h}, \quad y = h. & \text{(3.29c)} \end{cases}$$

Solving above system of equations, we get

$$\gamma(y) = \frac{-1}{h}\frac{\sinh(\frac{y\pi}{\mathbb{L}_0})}{\sinh(\frac{h\pi}{\mathbb{L}_0})}. \tag{3.30}$$

Substituting $\gamma(y)$ in (3.22), we have

$$\psi_1(\mathbb{X}, y) = -\frac{1}{h}\frac{\sinh(\frac{\pi y}{\mathbb{L}_0})}{\sinh(\frac{\pi h}{\mathbb{L}_0})}\cos(\frac{\mathbb{X}\pi}{\mathbb{L}_0}). \tag{3.31}$$

Using (3.31) in (3.21d), we have

$$-\frac{\pi}{h^2\mathbb{L}_0}\frac{\cosh(\frac{\pi h}{\mathbb{L}_0})}{\sinh(\frac{\pi h}{\mathbb{L}_0})}\cos(\frac{\mathbb{X}\pi}{\mathbb{L}_0}) + \cos(\frac{\mathbb{X}\pi}{\mathbb{L}_0}) = 0,$$

or

$$-\frac{\pi}{h^2\mathbb{L}_0}\frac{\cosh(\frac{\pi h}{\mathbb{L}_0})}{\sinh(\frac{\pi h}{\mathbb{L}_0})} + 1 = 0,$$

therefore

$$\tanh \frac{h\pi}{\mathbb{L}_0} = \frac{\pi}{h^2 \mathbb{L}_0}. \tag{3.32}$$

Relation (3.32) is called the dispersion relation and it is the same that we got in the previous chapter at (2.27). Putting $\tau = \dfrac{h\pi}{L_0}$ in above relation, we have

$$\frac{\tanh \tau}{\tau} = \frac{1}{h^3}, \tag{3.33}$$

since

$$\frac{\tanh \tau}{\tau} < 1,$$

then

$$\frac{1}{h^3} < 1,$$

which implies that h should be greater than 1. Thus in order to satisfy (3.32), we must choose the root of equation (3.19), which is greater than one. Therefore at $h > 1$, equation (3.33) has a solution τ and then L_0 is given by

$$L_0 = \frac{h\pi}{\tau}. \tag{3.34}$$

Thus we have found η_1, ψ_1 and $\mathbb{L}_0$ which are given by equations (3.23), (3.31) and (3.34) respectively.

3.3 Calculation of ψ_2, η_2 and $\mathbb{L}_1$

3.3.1 Boundary value problem for ψ_2, η_2 and $\mathbb{L}_1$

Using (3.18) and (3.31) in equation (3.13), we get

$$\psi_2(\mathbb{X}, h) = -\frac{\eta_2}{h} + \frac{h}{2} + \frac{h}{2}\cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}). \tag{3.35}$$

Using (3.18) and (3.31) in equation (3.16), we obtain

$$\frac{\pi^2}{4h^2\mathbb{L}_0^2}(1-\cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}))+(\frac{h^2}{4}-\frac{\pi^2}{2h^2\mathbb{L}_0^2})(1+\cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}))+\frac{1}{h}\psi_{2y}(\mathbb{X},h)+\eta_2=0. \tag{3.36}$$

Now combining equations (3.7), (3.10), (3.35) and (3.36), we formulate the following boundary value problem

$$\begin{cases} \psi_{2\mathbb{X}\mathbb{X}}+\psi_{2yy}=\dfrac{2\mathbb{L}_1}{h\mathbb{L}_0^3}\dfrac{\sinh(\frac{y\pi}{\mathbb{L}_0})}{\sinh(\frac{h\pi}{\mathbb{L}_0})}, \\ \\ \psi_2(\mathbb{X},0)=0, \quad \mathbb{X}\in[\mathbb{L}_0,\mathbb{L}_0] \\ \\ \psi_2(\mathbb{X},h)=-\dfrac{\eta_2}{h}+\dfrac{h}{2}+\dfrac{h}{2}\cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}), \quad \mathbb{X}\in[\mathbb{L}_0,\mathbb{L}_0] \\ \\ \dfrac{\pi^2}{4h^2\mathbb{L}_0^2}(1-\cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}))+(\dfrac{h^2}{4}-\dfrac{\pi^2}{2h^2\mathbb{L}_0^2})(1+\cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0})) \\ \\ +\dfrac{1}{h}\psi_{2y}(\mathbb{X},h)+\eta_2=0. \end{cases} \tag{3.37}$$

3.3.2 Solution to (3.37)

We seek a solution to the above boundary value problem in the following form

$$\psi_2(\mathbb{X},y)=\psi_2^{(P)}(\mathbb{X},y)+\psi_2^{(H)}(\mathbb{X},y), \tag{3.38}$$

where P and H are particular and homogeneous part of the solution respectively[3]. We are looking for the solution $\psi_2^{(P)}$ in the following form

$$\psi_2^{(P)}=A\sinh(\frac{\pi y}{\mathbb{L}_0}), \tag{3.39}$$

where A is an arbitrary constant. Substituting $\psi_2^{(P)}$ in first equation of (3.37), we have

$$A\frac{\pi^2}{\mathbb{L}_0^2}\sinh(\frac{y\pi}{\mathbb{L}_0})=0+2\frac{\mathbb{L}_1}{h\mathbb{L}_0^3}\frac{\sinh(\frac{y\pi}{\mathbb{L}_0})}{\sinh(\frac{h\pi}{\mathbb{L}_0})},$$

or

$$A = \frac{2\mathbb{L}_1}{h\pi^2\mathbb{L}_0 \sinh(\frac{\pi h}{\mathbb{L}_0})}.$$

Substituting the value of A in equation (3.39), we have

$$\psi_2^{(P)}(\mathbb{X}, y) = \frac{2\mathbb{L}_1 \sinh(\frac{\pi y}{\mathbb{L}_0})}{h\pi^2\mathbb{L}_0 \sinh(\frac{\pi h}{\mathbb{L}_0})}. \tag{3.40}$$

Function $\psi_2^{(H)}$ satisfies the first equation in (3.37) with zero right hand side

$$\psi_{2\mathbb{X}\mathbb{X}}^{(H)} + \psi_{2yy}^{(H)} = 0. \tag{3.41}$$

Using (3.38) and (3.40) in the 2nd equation of (3.37), we have

$$\psi_2^{(H)}(\mathbb{X}, 0) = 0. \tag{3.42}$$

Using (3.38) and (3.40) in the 3rd equation of (3.37), we have

$$\psi_2^{(H)}(\mathbb{X}, h) = -\frac{\eta_2}{h} + \frac{h}{2} + \frac{h}{2}\cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}) - \frac{2\mathbb{L}_1}{h\pi^2\mathbb{L}_0}. \tag{3.43}$$

Using (3.38) and (3.40) in the 4th equation of (3.37), we have

$$\frac{\pi^2}{4h^2\mathbb{L}_0^2}(1-\cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}))+(\frac{h^2}{4}-\frac{\pi^2}{2h^2\mathbb{L}_0^2})(1+\cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}))+\frac{1}{h}\psi_{2y}^{(H)}(\mathbb{X}, h)+\frac{2\mathbb{L}_1}{\pi^2\mathbb{L}_0}+\eta_2 = 0. \tag{3.44}$$

Now combining equations (3.41-3.44), we formulate the following homogeneous boundary value problem

$$\begin{cases} \psi_{2\mathbb{X}\mathbb{X}}^{(H)} + \psi_{2yy}^{(H)} = 0, \\ \\ \psi_2^{(H)}(\mathbb{X}, 0) = 0, \quad \mathbb{X} \in [-\mathbb{L}_0, \mathbb{L}_0], \\ \\ \psi_2^{(H)}(\mathbb{X}, h) = -\frac{\eta_2}{h} + \frac{h}{2} + \frac{h}{2}\cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}) - \frac{2\mathbb{L}_1}{h\pi^2\mathbb{L}_0}, \\ \\ \frac{\pi^2}{4h^2\mathbb{L}_0^2}(1-\cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0})) + (\frac{h^2}{4} - \frac{\pi^2}{2h^2\mathbb{L}_0^2})(1+\cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0})) \\ \\ +\frac{1}{h}\psi_{2y}^{(H)}(\mathbb{X}, h) + \frac{2\mathbb{L}_1}{\pi^2\mathbb{L}_0} + \eta_2 = 0. \end{cases} \tag{3.45}$$

We are looking for a solution $\psi_2^{(H)}$ in the following form

$$\psi_2^{(H)}(\mathbb{X}, y) = \psi^{(1)}(y) + \psi^{(2)} \cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}), \tag{3.46}$$

$$\eta_2 = \eta^{(1)} + \eta^{(2)} \cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}). \tag{3.47}$$

Using (3.46) in the first equation of (3.45), we have

$$\psi_{yy}^{(1)} + \left((\psi_{yy}^{(2)})^2 - \psi^{(2)}(\frac{4\pi^2}{\mathbb{L}_0^2}) \right) \cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}) = 0,$$

from which

$$(\psi_{yy}^{(2)})^2 - \psi^{(2)}(\frac{4\pi^2}{\mathbb{L}_0^2}) = 0, \tag{3.48}$$

and

$$\psi_{yy}^{(1)} = 0. \tag{3.49}$$

Using (3.46) in 2nd equation of (3.45), we have

$$\psi_2^{(H)}(\mathbb{X}, 0) = \psi^{(1)}(0) + \psi^{(2)} \cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}) = 0,$$

from which

$$\psi^{(2)}(0) = 0, \tag{3.50}$$

and

$$\psi^{(1)}(0) = 0. \tag{3.51}$$

Using (3.46) in 3rd equation of (3.45), we have

$$\psi^{(1)}(h) + \psi^{(2)} \cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}) = -\frac{1}{h}\left(\eta^{(1)} + \eta^{(2)} \cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}) \right) + \frac{h}{2} + \frac{h}{2} \cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}) - \frac{2\mathbb{L}_1}{h\pi^2\mathbb{L}_0},$$

from which

$$\psi^{(2)}(h) = \frac{h}{2} - \frac{\eta^{(2)}}{h}, \tag{3.52}$$

and

$$\psi^{(1)}(h) = -\frac{\eta^{(1)}}{h} + \frac{h}{2} - \frac{2\mathbb{L}_1}{h\pi^2\mathbb{L}_0}. \tag{3.53}$$

Using (3.46) in 4th equation of (3.45), we obtain

$$-\frac{\pi^2}{4h^2\mathbb{L}_0^2}+\frac{h^2}{4}-\frac{\pi^2}{2h^2\mathbb{L}_0^2}+\frac{1}{h}\psi_y^{(2)}(\mathbb{X},h)+\eta^{(2)}=0, \tag{3.54}$$

and

$$\frac{\pi^2}{4h^2\mathbb{L}_0^2}+(\frac{h^2}{4}-\frac{\pi^2}{2h^2\mathbb{L}_0^2})+\frac{2\mathbb{L}_1}{\pi^2\mathbb{L}_0}+\frac{1}{h}\psi_y^{(1)}(\mathbb{X},h)+\eta^{(1)}=0. \tag{3.55}$$

Now combining equations (3.48), (3.50), (3.52) and (3.54), we formulate the following boundary value problem for $\psi^{(2)}$

$$\begin{cases}\left(\psi_{yy}^{(2)}\right)^2-\psi^{(2)}(\frac{4\pi^2}{\mathbb{L}_0^2})=0,\\ \psi^{(2)}(0)=0,\\ \psi^{(2)}(h)=\frac{h}{2}-\frac{\eta^{(2)}}{h},\\ -\frac{\pi^2}{4h^2\mathbb{L}_0^2}+\frac{h^2}{4}-\frac{\pi^2}{2h^2\mathbb{L}_0^2}+\frac{1}{h}\psi_y^{(2)}(\mathbb{X},h)+\eta^{(2)}=0.\end{cases} \tag{3.56}$$

Solving (3.56) for $\psi^{(2)}$ with first two boundary conditions, we obtain

$$\psi^{(2)}=\left(-\frac{\eta^{(2)}}{h}+\frac{h}{2}\right)\frac{\sin h(\frac{2\pi y}{\mathbb{L}_0})}{\sinh(\frac{2\pi h}{\mathbb{L}_0})}. \tag{3.57}$$

Using (3.57) in forth equation of (3.56), we have

$$\frac{-3\pi^2}{4\mathbb{L}_0^2h^2}+\frac{h^2}{4}+\left(-\frac{\eta^{(2)}}{h}+\frac{h}{2}\right)h(1+\frac{\pi^2}{\mathbb{L}_0^2h^4})+\eta^{(2)}=0,$$

or

$$\eta^{(2)}=\frac{h^2}{4}\left(\frac{3\mathbb{L}_0^2h^4}{\pi^2}-1\right).$$

Now combining equations (3.49), (3.51), (3.53) and (3.55), we formulate the following boundary value problem for $\psi^{(1)}$

$$\begin{cases} \psi^{(1)}_{yy} = 0, \\ \\ \psi^{(1)}(0) = 0, \\ \\ \psi^{(1)}(h) = -\dfrac{\eta^{(1)}}{h} + \dfrac{h}{2} - \dfrac{2\mathbb{L}_1}{h\pi^2\mathbb{L}_0}, \\ \\ \dfrac{\pi^2}{4h^2\mathbb{L}_0^2} + (\dfrac{h^2}{4} - \dfrac{\pi^2}{2h^2\mathbb{L}_0^2}) + \dfrac{2\mathbb{L}_1}{\pi^2\mathbb{L}_0} + \dfrac{1}{h}\psi^{(1)}_y(\mathbb{X}, h) + \eta^{(1)} = 0. \end{cases} \tag{3.58}$$

We assume that $\eta^{(1)}$ is zero. Solving (3.58) for $\psi^{(1)}$ with first two boundary conditions, we obtain

$$\psi^{(1)} = -\frac{2\mathbb{L}_1 y}{h^2\pi^2\mathbb{L}_0} + \frac{y}{2}. \tag{3.59}$$

Using (3.59) in forth equation of (3.58), we have

$$\mathbb{L}_1 = \frac{\pi^2\mathbb{L}_0}{2(h^{-3} - \pi)}\left(\frac{h^2}{4} + \frac{1}{2h} - \frac{\pi^2}{4h^2\mathbb{L}_0^2}\right). \tag{3.60}$$

Using $\eta^{(1)}$ and $\eta^{(2)}$ in equation (3.47), we have

$$\eta_2(\mathbb{X}) = \frac{h^2}{4}\left(\frac{3\mathbb{L}_0^2h^4}{\pi^2} - 1\right)\cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}). \tag{3.61}$$

Using (3.57) and (3.59) in equation (3.46), we have

$$\psi_2^{(H)} = -\frac{2\mathbb{L}_1 y}{h^2\pi^2\mathbb{L}_0} + \frac{y}{2} + \left(-\frac{\eta^{(2)}}{h} + \frac{h}{2}\right)\frac{2\pi}{\mathbb{L}_0}\frac{\sinh(\frac{2\pi y}{\mathbb{L}_0})}{\sinh(\frac{2\pi h}{\mathbb{L}_0})}\cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}). \tag{3.62}$$

Using (3.40) and (3.62) in equation (3.38), we have

$$\psi_2(\mathbb{X}, y) = \frac{2\mathbb{L}_1}{h\pi^2\mathbb{L}_0\sinh(\frac{\pi h}{\mathbb{L}_0})}\sinh(\frac{\pi y}{\mathbb{L}_0}) - \frac{2\mathbb{L}_1 y}{h^2\pi^2\mathbb{L}_0} + \frac{y}{2} + \left(-\frac{\eta^{(2)}}{h} + \frac{h}{2}\right)\frac{\sinh(\frac{2\pi y}{\mathbb{L}_0})}{\sinh(\frac{2\pi h}{\mathbb{L}_0})}\cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0}). \tag{3.63}$$

Thus we have found $\mathbb{L}_1$, η_2 and ψ_2 which are given by equations (3.60), (3.61) and (3.63) respectively.

3.4 Flow Force

Since we assume that the Bernoulli constant is fixed. The flow force for steady waves can be deduced from expression for horizontal momentum flux plus pressure force, see [4]

$$\frac{3}{2}s = \frac{3}{2}r\eta(\mathbb{X},\epsilon) - \frac{1}{2}\eta(\mathbb{X},\epsilon)^2 + \int_0^{\eta(\mathbb{X},\epsilon)} \left(\frac{1}{2}\psi_y^2 - \frac{1}{2}(\frac{\mathbb{L}_0}{\mathbb{L}})^2\psi_{\mathbb{X}}^2\right) dy, \quad (3.64)$$

where

$$\mathbb{S} = \frac{3}{2}s, \quad (3.65)$$

$$\mathbb{R} = \frac{3}{2}r, \quad (3.66)$$

also see at figure (1.1), we have $\mathbb{X} = 0$. Now using (3.1), (3.65)-(3.66) in (3.64)

$$\mathbb{S} = \mathbb{R}\eta(\epsilon) - \frac{1}{2}\eta^2(\epsilon) + \int_0^{\eta(\epsilon)} \left(\frac{1}{2}(\psi_{0y} + \epsilon\psi_{1y} + \epsilon^2\psi_{2y})^2\right) dy + O(\epsilon^3),$$

using (3.2) and doing some simplification, we have

$$\mathbb{S} = \left(h^2 + \frac{1}{2h} - \frac{1}{2}h^2 + \frac{1}{2h}\right) + \epsilon\left(\frac{1}{h^2} - \frac{\pi}{\mathbb{L}_0 h^2 \frac{\pi}{\mathbb{L}_0}}\right) + \epsilon^2\left(\frac{1}{4}(\frac{3\mathbb{L}_0^2h^4}{\pi^2} - 1) + \right.$$
$$\left. \frac{\pi}{\mathbb{L}_0 h\sinh(\frac{\pi h}{\mathbb{L}_0})}(\frac{2\mathbb{L}_1}{\pi^2\mathbb{L}_0} - \frac{1}{h}) + \frac{\pi^2}{2\mathbb{L}_0^2 h\sinh^2(\frac{\pi h}{\mathbb{L}_0})} + (-\frac{\eta^{(2)}}{h} + \frac{h}{2})\frac{2\pi}{\mathbb{L}_0}\frac{1}{\sinh(\frac{2\pi h}{\mathbb{L}_0})}\right)$$
$$+O(\epsilon^3),$$

therefore

$$\mathbb{S} = \left(\frac{1}{h} + \frac{1}{2}h^2\right) + \epsilon^2\left(\frac{1}{4}(\frac{3\mathbb{L}_0^2h^4}{\pi^2} - 1) + \frac{\pi}{\mathbb{L}_0 h\sinh(\frac{\pi h}{\mathbb{L}_0})}(\frac{2\mathbb{L}_1}{\pi^2\mathbb{L}_0} - \frac{1}{h}) + \frac{\pi^2}{2\mathbb{L}_0^2 h\sinh^2(\frac{\pi h}{\mathbb{L}_0})}\right.$$
$$\left. + (-\frac{\eta^{(2)}}{h} + \frac{h}{2})\frac{2\pi}{\mathbb{L}_0}\frac{1}{\sinh(\frac{2\pi h}{\mathbb{L}_0})}\right) + O(\epsilon^3). \quad (3.67)$$

Using (3.2) at $\mathbb{X} = 0$, we have

$$\eta(\epsilon) = h + \epsilon + \epsilon^2\frac{h^2}{2}\left(\frac{3h^4\mathbb{L}_0^2}{\pi^2} - 1\right) + O(\epsilon^3), \quad (3.68)$$

The following figure shows the graph of flow force at different value of η which can obtain at different value of ϵ. We have chosen the following values for ϵ,

$$\epsilon = (0, \pm 0.01, \pm 0.02, ... \pm 0.08). \tag{3.69}$$

The following graph shows that the flow force $\mathbb{S}$ has minimum at $\eta = 1.5$.

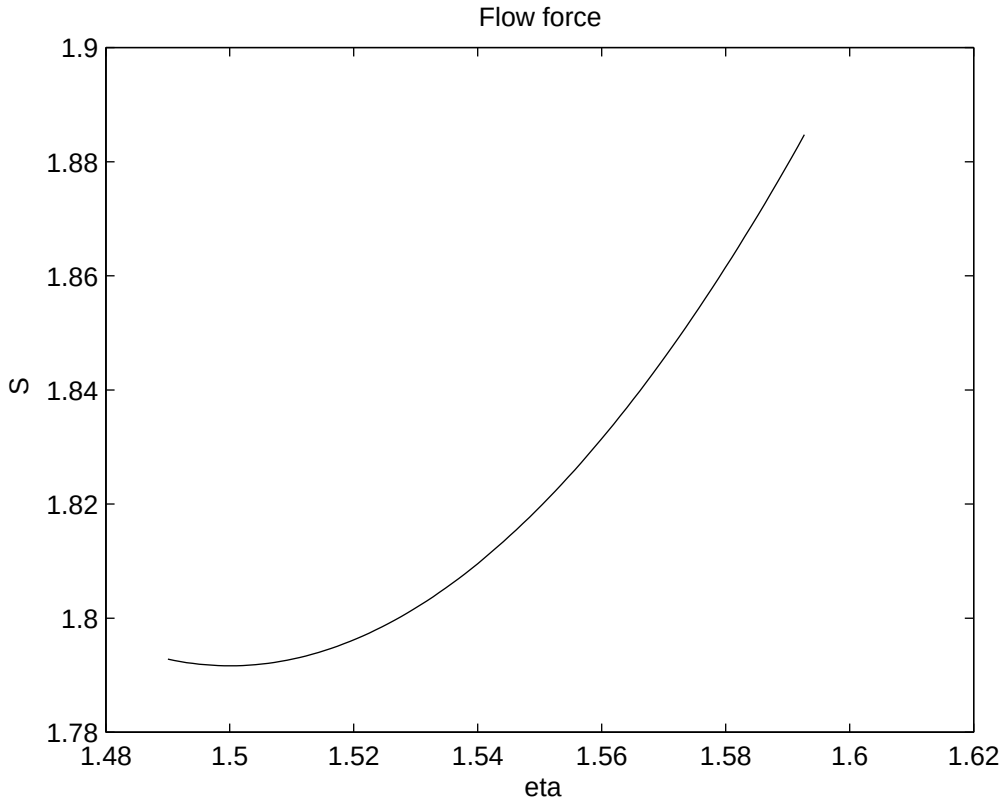


Figure 3.1: Overview of Flow Force

Using (3.19), we fixed the value of $\mathbb{R}$, which give us h,

$$\mathbb{R} = \frac{31}{18},$$

$$h = \frac{3}{2}.$$

Substituting the value of h in (3.34), we obtain the value for $\mathbb{L}_0$ which is 1.398. This $\mathbb{L}_0$ solves the dispersion relation at (3.32).

Results

First part

We have presented the periodic waves solution with the help of perturbation approach. The governing equations were formulated in terms of stream function see [5]. In the first part of the project, we assume that the period of the waves is fixed and stream function ψ, free surface η and pressure head $\mathbb{R}$ are unknown.
We are looking for unknown ψ, η and $\mathbb{R}$ in the form of (1.3)-(1.5) and asymptotically solving (1.2), we can write the following expressions for

$$\psi(x,y,\epsilon) = \frac{y}{h} + \epsilon\bigg(-\frac{1}{h}\frac{\sinh(\frac{\pi y}{L})}{\sinh(\frac{\pi h}{L})}\cos(\frac{x\pi}{L})\bigg) + \epsilon^2\bigg(\frac{3h}{4}(1-\frac{h^4L^2}{\pi^2}) \frac{\sinh(\frac{2\pi y}{L})}{\sinh(\frac{2\pi h}{L})}cos(\frac{2\pi x}{L}) + \frac{y}{2}\bigg) + O(\epsilon^3), \quad (3.70)$$

$$\eta(x,\epsilon) = h + \epsilon cos(\frac{\pi x}{L}) + \epsilon^2\bigg(\frac{h^2}{4}(\frac{3L^2h^4}{\pi^2}-1)\cos(\frac{2\pi x}{L})\bigg) + O(\epsilon^3), \quad (3.71)$$

and

$$\mathbb{R} = \mathbb{R}_0 + \epsilon^2\bigg(\frac{h^2}{4}(1-\tanh^2(\frac{\pi h}{L})) + \frac{1}{2h}\bigg) + O(\epsilon^3), \quad (3.72)$$

where

$$\mathbb{R}_0 = h + \frac{1}{2h^2} \quad (3.73)$$

$\mathbb{R}$ has minimum at $h = 1$, therefore (3.19) has solutions only when $\mathbb{R} \geq \frac{3}{2}$. We assume that $\mathbb{R} > \frac{3}{2}$, under this condition, the equation (3.19) has two

solutions. which implies that h should be greater than 1. Thus in order to satisfy (3.32), we must choose the root of equation (3.19), which is greater than one. Therefore at $h > 1$, equation (3.33) has a solution τ and then L_0 is given by

$$L = \frac{h\pi}{\tau}, \tag{3.74}$$

therefore L satisfies the dispersion relation (3.32).

Second part

We assume that the pressure head $\mathbb{R}$ is fixed and period $\mathbb{L}(\epsilon)$, free surface $\eta(\mathbb{X}, \epsilon)$ and stream function $\psi(\mathbb{X}, y, \epsilon)$ are unknown. We used the change of variable technique in the 2nd part to find the unknown ψ, η and $\mathbb{L}$.
We are looking for unknown ψ, η and $\mathbb{L}$ in the form of (3.1)-(3.3) and asymptotically solving (1.2), we can write the following expressions for

$$\psi(\mathbb{X}, y, \epsilon) = \frac{y}{h} + \epsilon\bigg(-\frac{1}{h}\frac{\sinh(\frac{\pi y}{\mathbb{L}_0})}{\sinh(\frac{\pi h}{\mathbb{L}_0})}\cos(\frac{\mathbb{X}\pi}{\mathbb{L}_0})\bigg) + \epsilon^2\bigg(\frac{2\mathbb{L}_1\sinh(\frac{\pi y}{\mathbb{L}_0})}{h\pi^2\mathbb{L}_0\sinh(\frac{\pi h}{\mathbb{L}_0})} -$$
$$\frac{2\mathbb{L}_1 y}{h^2\pi^2\mathbb{L}_0} + \frac{y}{2} + \bigg(-\frac{\eta^{(2)}}{h} + \frac{h}{2}\bigg)\frac{\sinh(\frac{2\pi y}{\mathbb{L}_0})}{\sinh(\frac{2\pi h}{\mathbb{L}_0})}\cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0})\bigg) + O(\epsilon^3), \tag{3.75}$$

$$\eta(\mathbb{X}, \epsilon) = h + \epsilon\bigg(\cos(\frac{\pi\mathbb{X}}{\mathbb{L}_0})\bigg) + \epsilon^2\bigg(\frac{h^2}{4}(\frac{3\mathbb{L}_0^2h^4}{\pi^2} - 1)\cos(\frac{2\pi\mathbb{X}}{\mathbb{L}_0})\bigg) + O(\epsilon^3), \tag{3.76}$$

and

$$\mathbb{L} = \mathbb{L}_0 + \epsilon\bigg(\frac{\pi^2\mathbb{L}_0}{2(h^{-3} - \pi)}(\frac{h^2}{4} + \frac{1}{2h} - \frac{\pi^2}{4h^2\mathbb{L}_0^2})\bigg) + O(\epsilon^2), \tag{3.77}$$

where

$$\mathbb{L}_0 = \frac{h\pi}{\tau} \tag{3.78}$$

which implies that h should be greater than 1. Thus in order to satisfy (3.32), we must choose the root of equation (3.19), which is greater than

one.
Last but not least the expression for flow force

$$\mathbb{S} = \left(\frac{1}{h}+\frac{1}{2}h^2\right)+\epsilon^2\left(\frac{1}{4}(\frac{3\mathbb{L}_0^2h^4}{\pi^2}-1)+\frac{\pi}{\mathbb{L}_0 h\sinh(\frac{\pi h}{\mathbb{L}_0})}(\frac{2\mathbb{L}_1}{\pi^2\mathbb{L}_0}-\frac{1}{h})+\frac{\pi^2}{2\mathbb{L}_0^2 h\sinh^2(\frac{\pi h}{\mathbb{L}_0})}\right.$$
$$\left.+(-\frac{\eta^{(2)}}{h}+\frac{h}{2})\frac{2\pi}{\mathbb{L}_0}\frac{1}{\sinh(\frac{2\pi h}{\mathbb{L}_0})}\right)+O(\epsilon^3). \quad (3.79)$$

Conclusion

In this thesis, bifurcation approach with a fixed period and Bernoulli constant for small periodic water waves in a finite depth channel is analyzed. The governing equations are formulated in term of stream function which satisfy the boundary value problem (1.2) in conjunction with free surface. Our aim to solve (1.2) asymptotically and find the relation for $\mathbb{R}$ and $\mathbb{S}$ which depend monotonically on height.

This thesis work is divided into two parts. In the first part, we solve the boundary value problem (1.2) asymptotically by fixing the period of the small periodic water waves. We have presented our results at (3.69)-(3.74) and compare with [2]. We observe, these results only exist for the value of h which is greater than one.

In the 2nd part we fixed the Bernoulli constant and obtain the results at (3.75)-(3.78). We found $\mathbb{R}$ at (3.19) and observe that it has two roots. We choose one of them which is greater than one, with this condition $\mathbb{L}_0$ satisfies the dispersion relation at (3.34). At the end we found the expression for flow force which depend monotonically on height, see at (3.67).

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